MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
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DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER - 2025**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Civil Engineering**	**Course Type**	**Core**
Course Name	**Environmental Studies**	**Course Code**	**CIV 1004**
Semester	**I**	**Max Marks**	**50**
Date	**28/11/2025**	**Duration**	**3 Hours**
SCHEME OF EVALUATION			

Q. No.	Question	Marks
1A	Distinguish between sustainable growth and environmental sustainability. Ans: Sustainable growth refers to the Development that meets the needs of the present without compromising the ability of future generation to meet their own needs. • In our world, development is primarily defined by GDP. Lesser developed economies have lower GDP. Developed economies have higher GDP. • For development to happen GDP must rise. But GDP is a measure of how much natural resources are consumed. So, resource consumption must increase for development to happen**. (2.5 Marks)** On the other hand, Environmental sustainability requires that we don't consume nonrenewable resources. In such a scenario, GDP should be close to zero. Environmental sustainability cannot be achieved as GDP grows as any consumption of such resources will reduce the environment as an entity. **(2.5 Marks)**	5
1B	What are the possible solutions for the Sundarbans crisis? Ans: Brief explanation on below points **(1 Mark each)** • Zoning according to vulnerability. • Planning, rehabilitation and relocation. • Embankments. • Creating opportunities that do not depend on nature. • Sustainable water management, including rainwater harvesting and desalination units.	5

2A	Describe on how citizen science has helped in solving wildlife conflicts and conservation of Amur falcon. Ans: **(2.5 Marks each)** Citizens living near forests often act as the first responders by reporting wildlife sightings, crop raids, or conflict incidents through mobile apps and community networks. • Wild seve is a mobile technology-based platform where a farmer can call a toll-free number to report a wildlife-conflict related incident. Trained personnel then physically come to help the farmer in filing compensation claims and other due processes to get fair compensation for their damage from the government. Amur Falcon is a small pigeon-sized raptor. It has one of the longest migration routes in the world, covering almost 22000km in a single year. They are heavily hunted during their stop at Northeast India. • ebird, an online citizen science portal helped in collecting data on the observations of birds • Ending Hunting Through Local Engagement • Eco-tourism and Incentives	**5**
2B	Define the ecological footprint and explain why it is considered a more comprehensive metric for environmental impact? Ans: Ecological Footprint' is the ecological services that a given population requires to produce the natural resources it consumes and to assimilate waste and carbon emissions generated. An ecological deficit occurs when the Ecological Footprint of a population exceeds the biocapacity of the area available to that population. **(2 Marks)** It's a comprehensive metric for environmental impact as **(3 Marks)** • It includes not only the resources consumed but also the ecological area needed to absorb waste, especially CO_2 emissions, giving a fuller picture of human pressure on the environment. • It highlights how lifestyle, economic growth, and consumption habits affect the environment • It compares ecological footprint with the planet's biocapacity, helping identify ecological deficits or sustainability levels. A national ecological deficit means that the nation is importing biocapacity through trade, liquidating national ecological assets or emitting carbon dioxide waste into the atmosphere. An ecological reserve exists when the biocapacity of a region exceeds its population's Ecological Footprint. • By expressing impacts in terms of "global hectares," it converts different types of resource use into a single comparable metric.	**5**

3A	How does air pollution contribute to climate change and acid rain? Ans: **Five points (1 Marks each)** • Greenhouse gas emissions such as carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O) trap heat in the atmosphere, leading to global warming. • Air pollution intensifies temperature rise, extreme weather events, sea-level rise, and climate disruptions. • Sulphur dioxide (SO_2) released from coal-fired power plants, industries, and volcanic activity reacts with oxygen and water vapour to form sulphuric acid (H_2SO_4). • Nitrogen oxides (NOx) from vehicles, industries, and combustion processes react to form nitric acid (HNO_3). • These acids fall to the Earth as acid rain, affecting soil, water bodies, forests, and buildings.	**5**
3B	Explain the effect of nutrient loading and salinization on the water quality. Ans: **2.5 Marks each** Nutrient loading refers to the excessive input of nitrogen and phosphorus into water bodies from fertilizers, sewage, agricultural runoff, and industrial effluents popularly known as Eutrophication. Excess nutrients stimulate rapid algal growth (algal blooms). When algae die and decompose, oxygen levels drop, leading to low oxygen conditions resulting in loss of aquatic biodiversity. High salt concentration makes water unsuitable for drinking, agriculture, and industrial use. Saline water used for irrigation leads to salt accumulation in soils, reducing fertility and crop productivity. It corrodes pipelines and treatment equipment. Increased levels of chloride and sodium change the natural ionic balance of water.	**5**
4A	Discuss the legal framework for water resource management in India with reference to the Easement Rights and the River Boards Act. Ans: **Act objective- 2 Marks, description- 3 Marks** The Easements Act 1882 provides the foundation for individual and community water rights, ensures fair access, and prevents conflicts related to water use at the local level. • Owners of land have the right to draw and use sub-surface water, provided it does not interfere with legally established rights of neighbours • A person cannot Divert the natural course of water The River Boards Act was enacted to promote inter-state cooperation in managing and developing river systems. • Regulation and development of inter-state rivers • Coordinating irrigation, hydropower, flood control and navigation • Advising states on water resource planning	**5**

	• Ensuring equitable water allocation	
4B	Discuss how the Indian law ensures safety from hazardous Industries and immediate relief to affected person in case of any accident. Ans: **Listing the acts- 2 Marks, Explanation- 3 Marks** • 1989 - The Manufacture, Storage, and Import of Hazardous Chemicals Defines the terms used in this context, and sets up an authority to inspect, once a year, the industrial activity connected with hazardous chemicals and isolated storage facilities. • 1992 - The Public Liability Insurance Act and Rules and Amendment Drawn up for the purpose of providing immediate relief to the persons affected by accident while handling any hazardous substance. • 1995 - The National Environmental Tribunal Act Created to award compensation for damages to persons, property, and the environment arising from any activity involving hazardous substances. • Environment (Protection) Act, 1986 Empowers the Central Government to set standards for emissions, effluents, hazardous substance handling, storage, and transportation. Enable authorities to shut down or regulate hazardous industries that violate norms	**5**
5A	Effective watershed management can reduce the severity of floods in urban areas. Justify the statement. Ans: Watershed management focuses on conserving soil, water, and vegetation within a drainage basin. In urban areas, where rapid runoff, impervious surfaces, and poor drainage increase flood risks, effective watershed management plays a crucial role in mitigating floods. **(1 Marks)** • It helps increase groundwater recharge, reducing the volume and velocity of surface runoff that causes flooding. • Watershed practices such as contour bunding, afforestation, and check dams reduce sediment load and maintain drainage efficiency. • Temporarily store stormwater during heavy rainfall and release it slowly, preventing sudden surges in stormwater drains. • Helps with coordinated stormwater design, and synchronized reservoir releases. **(4 Marks)**	**5**

5B	Discuss the geophysical and hydrological types of natural disasters. Ans: Discussion on any two under each **(2.5 Marks each)** Geophysical • Earthquakes • Tsunamis • Volcanic Eruptions • Landslides Hydrological • Floods • Drought • Cyclones	**5**